

SATHSARANI SAUBHAGYA

Software Engineering Undergraduate | Full-Stack Developer

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[GitHub](#) | [LinkedIn](#) | [Portfolio](#)

PROFESSIONAL PROFILE

Motivated Information Technology undergraduate with hands-on experience in Full-Stack Web Development and Flutter Mobile Application Development. Proficient in developing responsive web applications, cross-platform mobile applications, and database-driven systems using Flutter, Firebase, JavaScript, PHP, Java, MySQL, and Git/GitHub for version control. Familiar with Software Quality Assurance (QA) principles, including manual testing, test case preparation, bug reporting, Agile methodologies, and the Software Development Life Cycle (SDLC). Passionate about developing innovative software solutions and continuously expanding my technical expertise. A quick learner with strong analytical thinking, problem-solving, communication, and teamwork skills, seeking a Software Engineering, AI, or QA Internship where I can apply my technical skills while contributing to real-world projects. Particularly interested in Full-Stack Development, Artificial Intelligence, Mobile Application Development, and Software Quality Assurance

TECHNICAL SKILLS

- **Programming Languages:** - JavaScript, Dart, Java, Python, C#, TypeScript
- **Frameworks & Libraries:** - Flutter, React.js, Express.js
- **Frontend Development:** - HTML5, CSS3, JavaScript, React.js (Basic)
- **Backend Development:** - PHP, Node.js, REST APIs, Express.js
- **Mobile application development:** - flutter, Dart
- **Databases:** - MongoDB, MySQL, Firebase Firestore, Firebase Realtime Database
- **Tools & Platforms:** - Git, GitHub, Postman, Figma, Visual Studio Code, NetBeans, Firebase Console

PROFESSIONAL EXPERIENCE

QA Hackathon Participant & Industry Visit | 99x Technology | 2026

- Participated in a Quality Assurance (QA) Workshop and Hackathon, collaborating with team members to perform software testing activities, identify defects, and improve application quality through structured testing approaches.
- Gained practical exposure to Software Quality Assurance (SQA), Software Testing Life Cycle (STLC), Agile methodologies, and industry best practices through technical sessions and hands-on activities.
- Developed analytical thinking, problem-solving, communication, and teamwork skills by participating in QA-focused challenges and interacting with software engineering professionals during the industry visit.

Sales Assistant | Alankulama (Pvt) Ltd | 2024

- Managed inventory operations by receiving, recording, organizing, and distributing stock across multiple product categories, including bakery products, vegetables, meat products, seafood, supermarket items, pastries, household goods, and catering supplies, ensuring accurate inventory control and product availability.
- Maintained purchase records, invoices, inventory documentation, and stock reports with a high level of accuracy to support efficient business operations.
- Updated and managed product information, pricing, and stock records using a Point of Sale (POS) system, ensuring accurate sales transactions and real-time inventory updates.

- Conducted market and competitor price analysis to determine competitive pricing for bulk food items, fresh produce, and retail products based on daily market trends.
- Coordinated with suppliers and internal departments to ensure timely stock replenishment, efficient inventory distribution, and uninterrupted business operations.
- Strengthened analytical thinking, data management, problem-solving, communication, and organizational skills while working in a fast-paced retail environment.

TECHNICAL PROJECTS

EduQuiz-App - G.C.E. O/L Mobile Learning Application

Self-Directed Project | Flutter + Dart + Firebase (Firestore, Authentication, Storage)

- Developed a cross-platform mobile learning application for G.C.E. O/L students using Flutter and Dart.
- Built a dynamic quiz system with subject selection, timed exams, automatic submission, and instant result generation.
- Integrated Firebase Authentication, Cloud Firestore, and Firebase Storage for secure authentication and real-time data management.
- Implemented PDF generation and offline access to downloaded quiz papers.
- Designed a scalable and user-friendly educational platform to enhance students' learning experience.
- GitHub: <https://github.com/sSaubhagya12/quiz-app>

School Staff Management System - Full-Stack School Administration & Staff Portal

Self-Directed Project | React.js | Node.js | Express | JavaScript | HTML5/CSS3 / MongoDB

- Designed and built a complete web-based management system to automate school administrative operations and staff tracking.
- Featured QR-code-based attendance scanning, role-based dashboards (Admin & Staff), and real-time announcement notifications.
- Managed automated staff timetables, payroll calculations, personal details registry, and complaint submission systems.
- Created robust RESTful APIs using Node.js & Express with a custom data management layer for fast request processing.
- GitHub: https://github.com/sSaubhagya12/School_Staff_ManagSys.git

SmartNav-App - AI & Sensor-Based Indoor Assistive Navigation Mobile Application

Self-Directed Project | React Native + Expo + TypeScript + Motion Sensors (PDR) + Voice Recognition

- Developed a cross-platform mobile navigation application designed to assist users with real-time indoor navigation using React Native, Expo, and TypeScript.
- Built a custom Pedestrian Dead Reckoning (PDR) Engine leveraging accelerometer and magnetometer sensor data for real-time step tracking, heading orientation, and indoor relative position calculation (X, Y coordinate mapping).
- Integrated Voice Control and Text-to-Speech (TTS) navigation guidance using expo-speech-recognition and expo-speech for hands-free interaction.
- Implemented real-time safety algorithms for Fall Detection and Obstacle Warning Systems based on threshold peak analysis.
- Engineered multi-sensory feedback utilizing dynamic audio soundscapes (expo-av) and tactile haptic alerts (expo-haptics) for an enhanced accessibility user experience.
- Integrated camera vision modules, custom user profile setup, and an administrative management dashboard.
- GitHub: <https://github.com/sSaubhagya12/smartnav-app>

EDUCATION

Higher National Diploma in Information Technology <i>Sri Lanka Institute of Advanced Technological Education (SLIATE)</i>	Expected 2027
G.C.E. Advanced Level – Technology Stream Results: 2 C, 1 S	2022 / 2023
G.C.E Ordinary Level Results: 3 A, 3 B, 3 C	2018

CERTIFICATIONS

- C for Beginners – – Great Learning Academy (2024)
- User Interface Design with Figma – Alison (2025)
- Introduction to Figma – SimpliLearn (2025)
- Introduction to JavaScript – Great Learning Academy (2024)

SOFT SKILLS & LANGUAGES

Soft Skills:

- Problem Solving
- Analytical Thinking
- Team Collaboration
- Communication Skills
- Time Management
- Self-Learning
- Adaptability
- Attention to Detail
- Critical Thinking

Languages:

- Sinhala (Native Proficiency)
- English (Professional Working Proficiency)

REFERENCES

Available upon request.